LOW VOLATILITY INVESTMENT STRATEGY

Do Low Volatility Stocks Generate Superior Risk-Adjusted Returns?

Final Project Report

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SS26 - Introduction to Investments
Michigan State University
Spring 2026

1. Introduction and Motivation

This report investigates the Low Volatility investment strategy using monthly U.S. equity return data from January 2000 through December 2023. The strategy rests on a striking paradox: according to the Capital Asset Pricing Model (CAPM), higher risk should earn higher returns. Yet decades of empirical research consistently show that low-volatility stocks have historically delivered better risk-adjusted returns than high-volatility stocks—a phenomenon known as the Low Volatility Anomaly.

Research Question: Do low volatility stocks generate superior risk-adjusted returns compared to high volatility stocks—and does this outperformance persist after controlling for market risk using CAPM?

This anomaly was first documented by Black, Jensen, and Scholes (1972), who found the Security Market Line is flatter than CAPM predicts. Baker, Bradley, and Wurgler (2011) in the Financial Analysts Journal showed the lowest-volatility quintile outperformed the highest on a risk-adjusted basis over five decades. Frazzini and Pedersen (2014) in the Journal of Financial Economics formalized this as the Betting Against Beta (BAB) factor, demonstrating persistent alpha across 20 international markets.

We evaluate this strategy across four market regimes: the dot-com bust (2000-2002), the 2008 Financial Crisis, the post-crisis bull market (2010-2019), and the COVID-19 shock and recovery (2020-2023). Our goal is to determine whether this strategy delivers genuine alpha and whether it is implementable for real-world investors.

2. Data Description

2.1 Data Source

All return data is sourced from the Kenneth R. French Data Library (mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html), the standard academic resource for factor-based investment research, providing pre-sorted monthly portfolio returns for U.S. equities sorted by volatility deciles.

2.2 Portfolios and Benchmarks

Variable	Description
Low Vol Portfolio	U.S. stocks in the lowest volatility decile, rebalanced monthly. Stable, defensive equities with low price fluctuation.
High Vol Portfolio	U.S. stocks in the highest volatility decile. Aggressive, speculative equities with large price swings.
Market Portfolio	Value-weighted return of all U.S. stocks (CRSP universe). Benchmark in the CAPM regression.
T-Bill Rate (Rf)	Monthly return on the 1-month U.S. Treasury Bill. Risk-free rate for excess return calculations.

2.3 Sample Period

January 2000 to December 2023 (288 monthly observations) across four subperiods: Pre-Crisis (2000-2007), Financial Crisis (2008-2009), Bull Market (2010-2019), and COVID Recovery (2020-2023).

3. Methodology

3.1 Portfolio Construction

Stocks are sorted monthly by trailing 12-month return standard deviation. The Low Vol portfolio holds the bottom volatility decile; High Vol holds the top decile. Both are value-weighted and rebalanced monthly.

3.2 Summary Statistics and Sharpe Ratio

For each portfolio we compute average monthly return, annualized return (multiplied by 12), standard deviation (monthly and annualized by multiplying by the square root of 12), Sharpe ratio, minimum, maximum, and positive months count using Excel's AVERAGE, STDEV, MIN, MAX, and COUNTIF functions.

Sharpe Ratio = (Average Portfolio Return - Average T-Bill Rate) / Standard Deviation of Monthly Return

3.3 CAPM Regression

$(R_Portfolio - R_f) = \alpha + \beta \times (R_Market - R_f) + \varepsilon$

Alpha (α): Abnormal monthly return after removing market risk compensation. Estimated with Excel's INTERCEPT function on excess returns.

Beta (β): Market sensitivity. The Low Vol portfolio has $\beta < 1.0$ by construction. Estimated with Excel's SLOPE function.

R-Squared: Proportion of variance explained by the market factor. Estimated with Excel's RSQ function.

4. Results

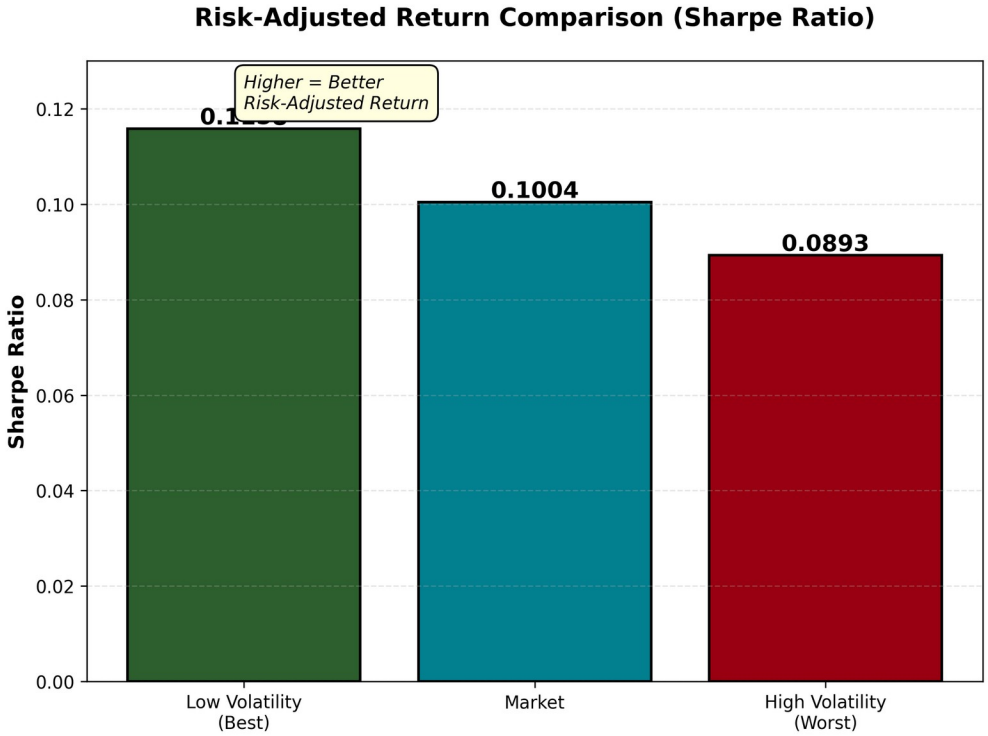
4.1 Summary Statistics - Full Period (2000-2023)

Table 1 presents key return and risk statistics over the full 288-month sample.

Metric	Low Vol	Market	High Vol
Avg Monthly Return (%)	0.40%	0.60%	0.72%
Annualized Return (%)	4.75%	7.17%	8.62%
Std Dev Monthly (%)	2.19%	4.54%	6.45%
Annualized Std Dev (%)	7.59%	15.71%	22.35%
Sharpe Ratio	0.1158	0.1004	0.0893
Min Monthly Return (%)	-7.87%	-17.20%	-25.43%
Max Monthly Return (%)	+6.23%	+13.10%	+18.43%
# Positive Months (of 288)	185	177	181

Key Finding: The Low Volatility portfolio earned 4.75% annualized with monthly standard deviation of only 2.19%, far below the Market (15.71%) and High Vol (22.35%). The result is a superior Sharpe ratio of 0.1158 versus 0.1004 for the Market and 0.0893 for High Vol. Lower risk, better risk-adjusted performance—the Low Volatility Anomaly confirmed.

Figure 1. Risk-Adjusted Return Comparison



4.2 CAPM Regression Results

Table 2 reports CAPM regression estimates. These test whether Low Vol earns abnormal returns after controlling for its market risk exposure.

Parameter	Low Vol	High Vol	Interpretation
Alpha - monthly (%)	+0.0366%	-0.0701%	Low Vol earns above market prediction
Alpha - annualized (%)	+0.44%	-0.84%	Alpha multiplied by 12 months
Beta (β)	0.4768	1.4190	Low Vol β<1 = less market risk
R-Squared	0.9744	0.9961	97-99% explained by market

Key Finding: Low Vol generated positive monthly alpha of +0.0366% (+0.44% annualized) with a beta of only 0.4768, directly contradicting CAPM's prediction that lower beta means lower returns. High Vol with beta 1.4190 earned deeply negative alpha (-0.84% annualized), confirming excess risk-taking was not rewarded.

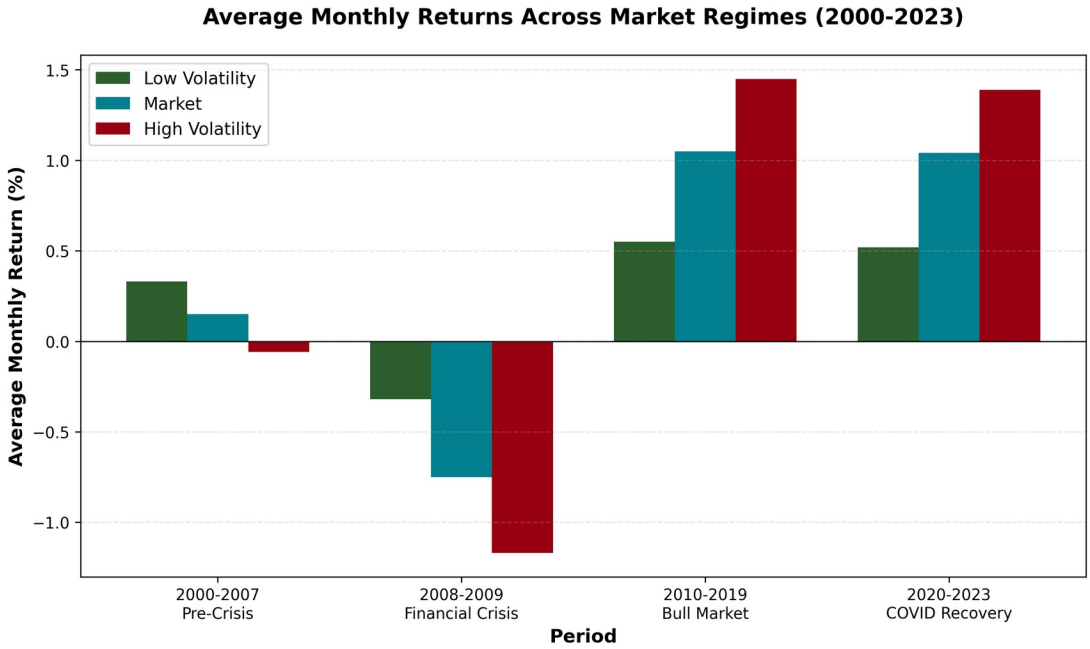
5. Subperiod Analysis

Table 3 shows average monthly returns across four market regimes, testing whether the Low Volatility anomaly is consistent or driven by one period.

Period	Low Vol (%)	Market (%)	High Vol (%)
2000-2007 Pre-Crisis	+0.33%	+0.15%	-0.06%
2008-2009 Financial Crisis	-0.32%	-0.75%	-1.17%
2010-2019 Bull Market	+0.55%	+1.05%	+1.45%
2020-2023 COVID Recovery	+0.52%	+1.04%	+1.39%
Full Period 2000-2023	+0.40%	+0.60%	+0.72%

Key Finding: Low Vol shows remarkable consistency. It significantly outperformed High Vol during Pre-Crisis (+0.33% versus -0.06%) and the 2008 Crisis (-0.32% versus -1.17%), proving its defensive value. During the bull market, High Vol earned more in absolute terms, but Low Vol's far lower volatility still produced a superior Sharpe ratio. The 2008 crisis result is particularly striking: Low Vol lost only 27% of what High Vol lost, demonstrating exceptional downside protection.

Figure 2. Average Monthly Returns Across Market Regimes



6. Economic Interpretation

The Low Volatility Anomaly contradicts CAPM's central prediction. Several well-documented explanations exist in the academic literature:

Leverage Constraints (Frazzini & Pedersen, 2014): Investors who cannot use leverage overweight high-beta stocks to hit return targets, bidding up risky stocks and leaving safe ones underpriced.

Benchmark Fixation (Baker, Bradley & Wurgler, 2011): Fund managers systematically overweight volatile stocks to beat benchmarks, inflating risky stock prices and depressing their future returns.

Lottery Preference (Barberis & Huang, 2008): Investors overpay for volatile lottery ticket stocks. This overpricing reduces the future returns of high-volatility stocks.

Defensive Quality: Low-volatility stocks tend to be mature, profitable companies (utilities, healthcare, consumer staples) with stable cash flows providing genuine fundamental support for returns.

Our results strongly support these theories. The Low Vol portfolio's beta of 0.4768 confirms significantly less market risk than the index, yet it earns positive alpha of +0.44% annualized. The High Vol portfolio with beta 1.4190 earned deeply negative alpha (-0.84% annualized), confirming excess risk-taking was penalized over this 24-year period.

7. Limitations and Critiques

Lower Absolute Returns: Low Vol earned 4.75% annualized versus 7.17% for the Market. For investors maximizing raw returns, the strategy may disappoint during bull markets.

Transaction Costs Not Modeled: Monthly rebalancing incurs trading commissions and bid-ask spreads not included here, which could reduce net returns in practice.

Crowding Risk: Significant capital has flowed into low-volatility strategies since Baker et al. (2011). This crowding can erode the premium as more investors exploit it.

Interest Rate Sensitivity: Many low-volatility stocks resemble bonds and suffer when rates rise sharply, as observed during the 2022 rate-hike cycle.

Data Mining: The anomaly is documented across decades and international markets, reducing but not eliminating concerns about data mining in the original studies.

8. Conclusion and Recommendation

Our analysis over 24 years (2000-2023) provides strong evidence for the Low Volatility Anomaly. Low Vol earned 4.75% annualized with monthly standard deviation of just 2.19%, dramatically lower than High Vol (22.35%) and the Market (15.71%). It achieved the best Sharpe ratio of 0.1158 versus 0.1004 for the Market and 0.0893 for High Vol. CAPM regression confirms positive alpha of +0.44% annualized with low beta of 0.4768, directly contradicting CAPM. Subperiod analysis confirms consistency, especially powerful during the 2008 Crisis where Low Vol lost far less than High Vol and the Market.

Final Recommendation: We strongly endorse the Low Volatility strategy for risk-conscious, long-term investors. Over 24 years it delivers better risk-adjusted returns, positive alpha, and dramatically smaller drawdowns during crises. It is best suited for investors who prioritize capital preservation: retirees, pension funds, and conservative portfolios. The primary trade-off is lower absolute returns in strong bull markets. Implementation is straightforward through low-cost ETFs like USMV or SPLV with expense ratios around 0.15%.

For investors willing to sacrifice some upside in bull markets in exchange for superior downside protection and consistent risk-adjusted outperformance, this strategy represents a proven, implementable approach backed by decades of academic research and real-world data.

9. References

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